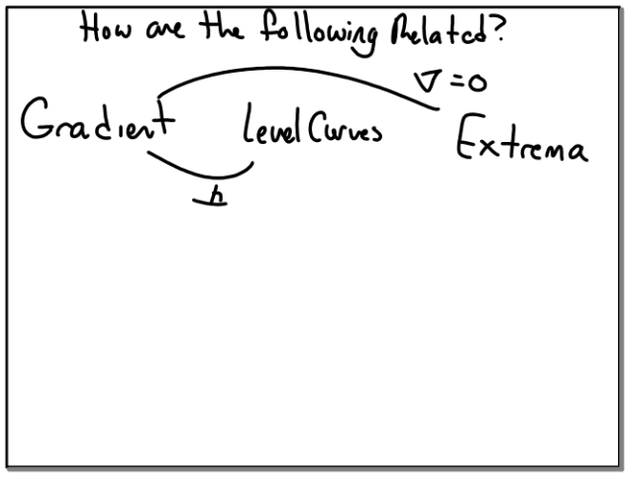
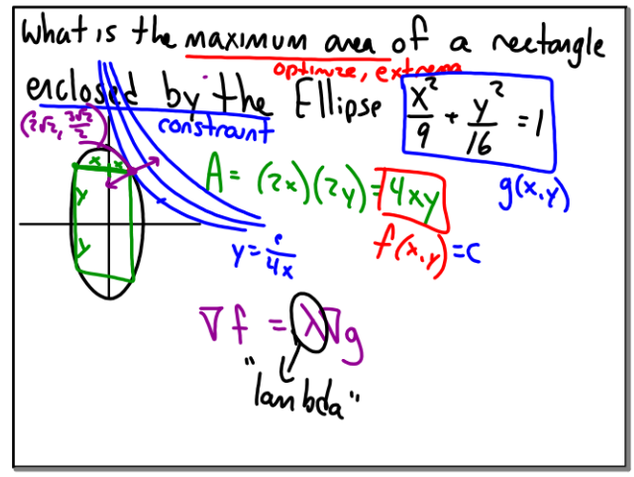
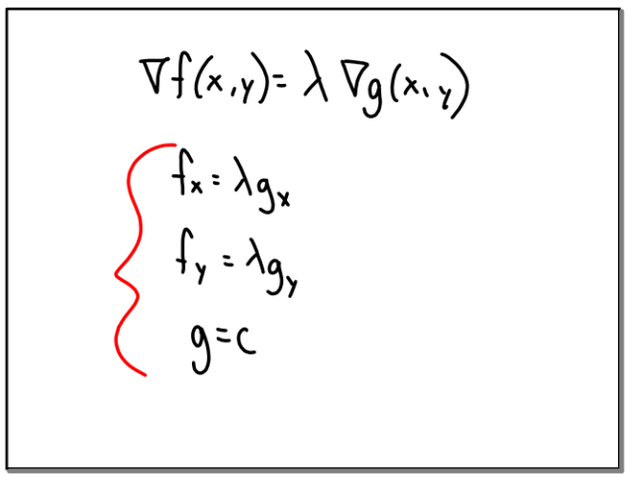


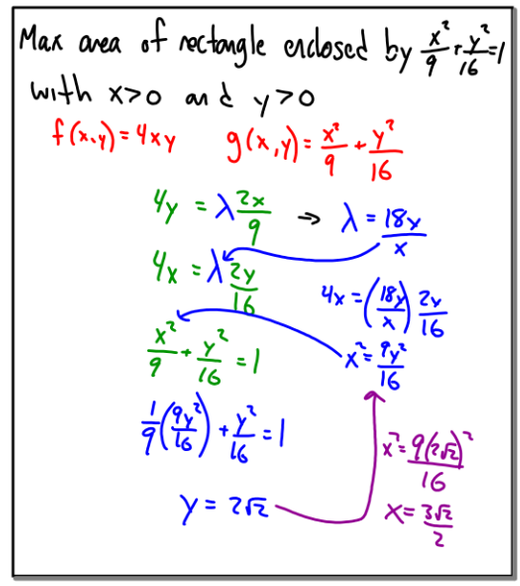
How the ideas connect: the gradient is perpendicular (⊥) to level curves, and at an extremum with no constraint, ∇f = 0. With a constraint, the next slides use that perpendicularity instead.



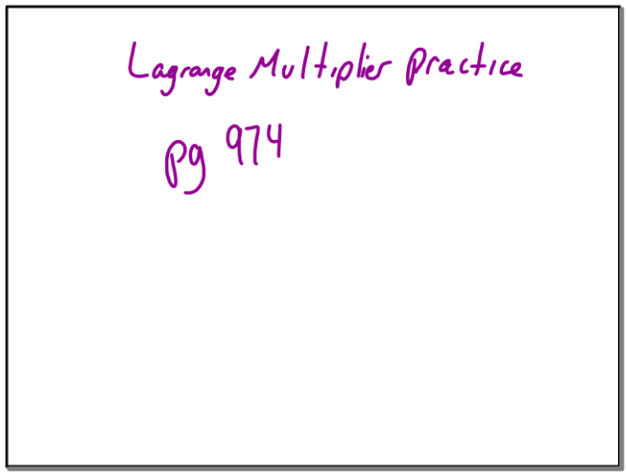
Maximize A = (2x)(2y) = 4xy for a rectangle inside $x^2/9 + y^2/16 = 1$ (the constraint g). The level curves 4xy = c are hyperbolas; the best one just touches the ellipse. There the normals line up: ∇f = λ∇g (λ is "lambda").



Lagrange multipliers: to optimize f subject to g = c, solve $f_x = \lambda g_x$, $f_y = \lambda g_y$ and g = c together. Then plug every solution into f; the largest value is the max and the smallest is the min.



Process: 4y = λ(2x/9) gives λ = 18y/x; put it into 4x = λ(2y/16) to get $x^2 = 9y^2/16$. Substitute into the ellipse: $y^2/8 = 1$, so $y = 2\sqrt{2}$ and $x = 3\sqrt{2}/2$. Max area = 4xy = $4(3\sqrt{2}/2)(2\sqrt{2}) = 24$.



Practice problems for Lagrange multipliers are on textbook page 974.

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